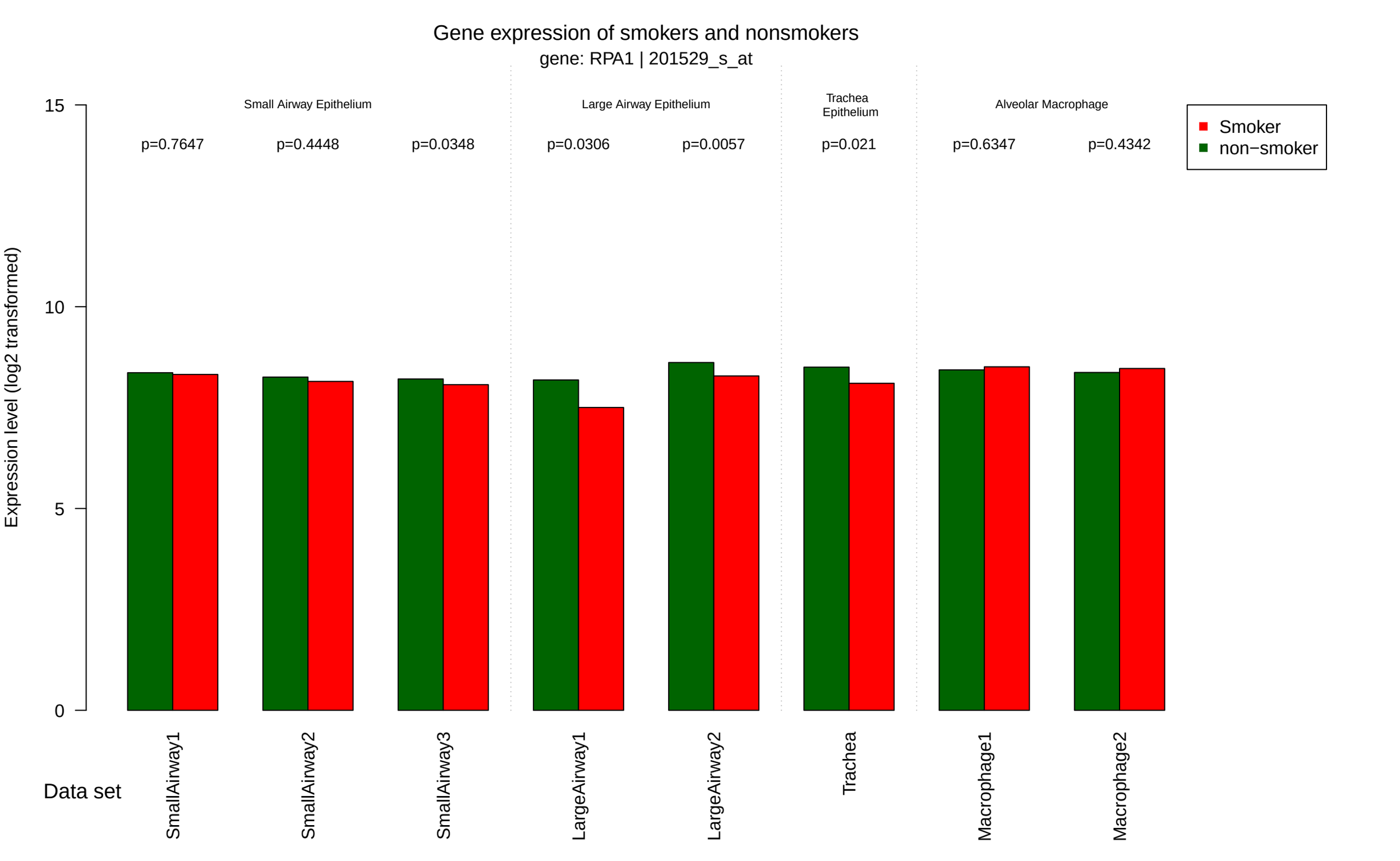
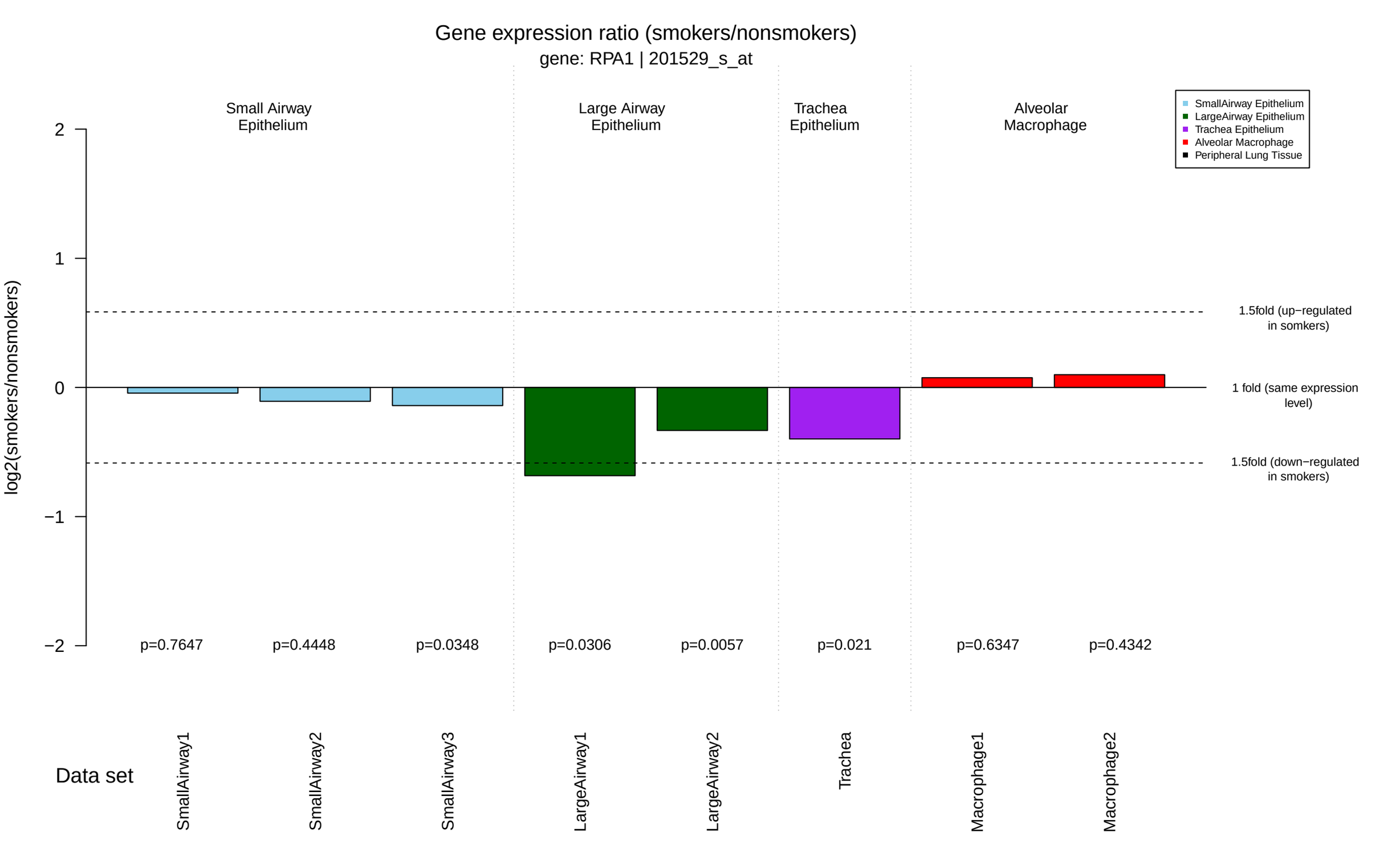
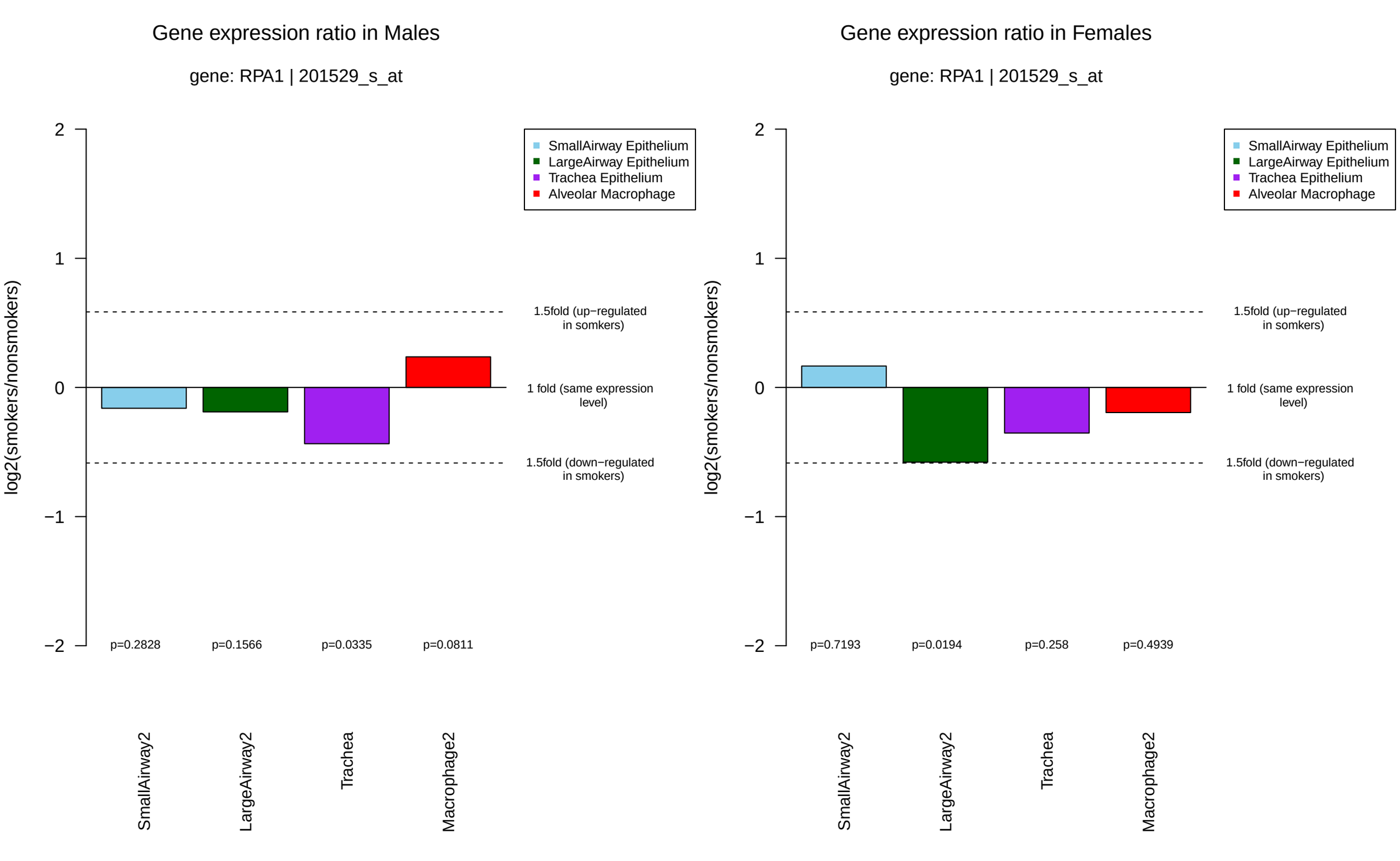
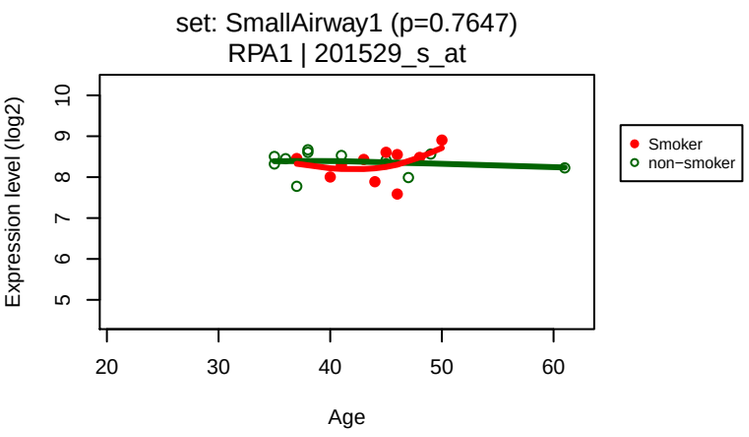
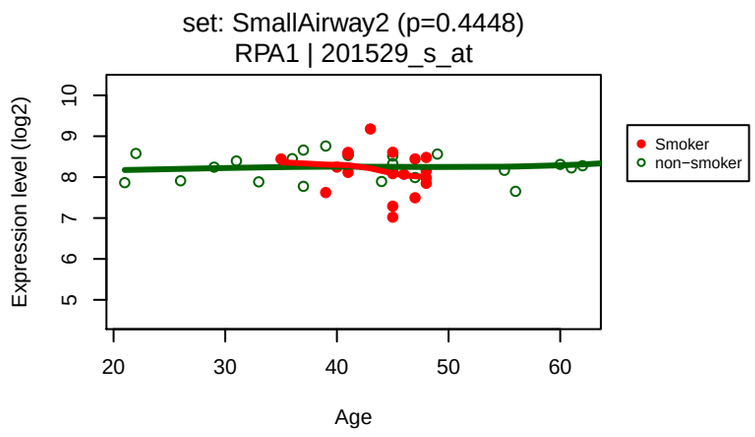
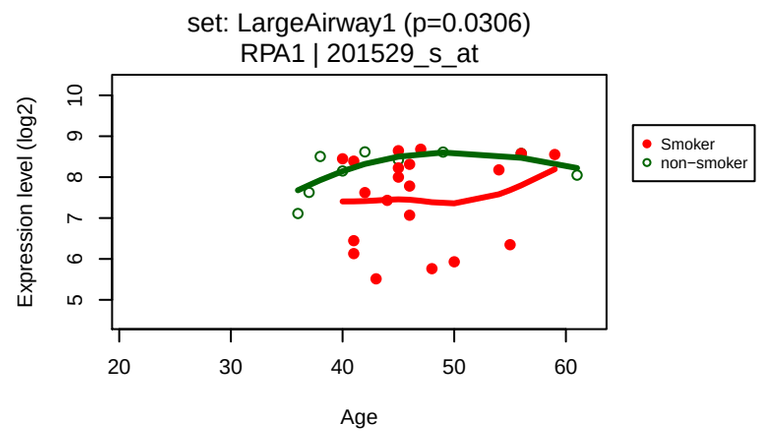

Gene expression of smokers and nonsmokers
gene: RPA1 | 201529_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7647
p=0.4448
p=0.0348
p=0.0306
p=0.0057
p=0.021
p=0.6347
p=0.4342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

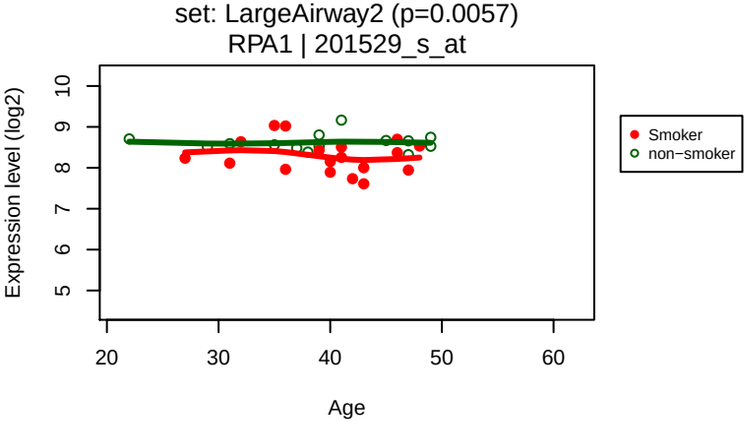

Gene expression ratio (smokers/nonsmokers)
gene: RPA1 | 201529_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7647
p=0.4448
p=0.0348
p=0.0306
p=0.0057
p=0.021
p=0.6347
p=0.4342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

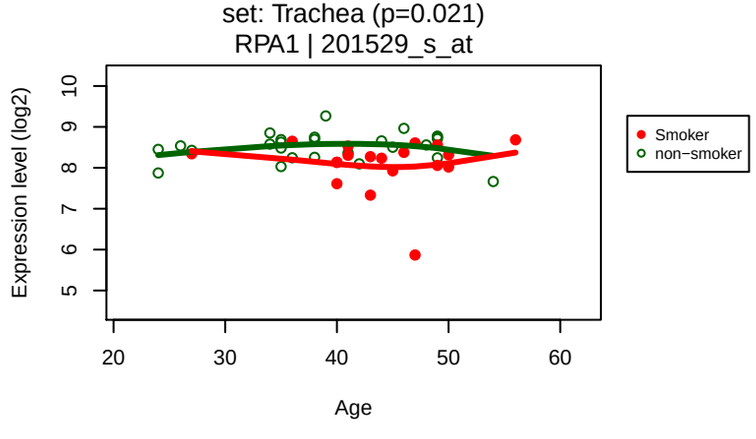

Gene expression ratio in Males
gene: RPA1 | 201529_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2828
p=0.1566
p=0.0335
p=0.0811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPA1 | 201529_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7193
p=0.0194
p=0.258
p=0.4939
SmallAirway2
LargeAirway2
Trachea
Macrophage2

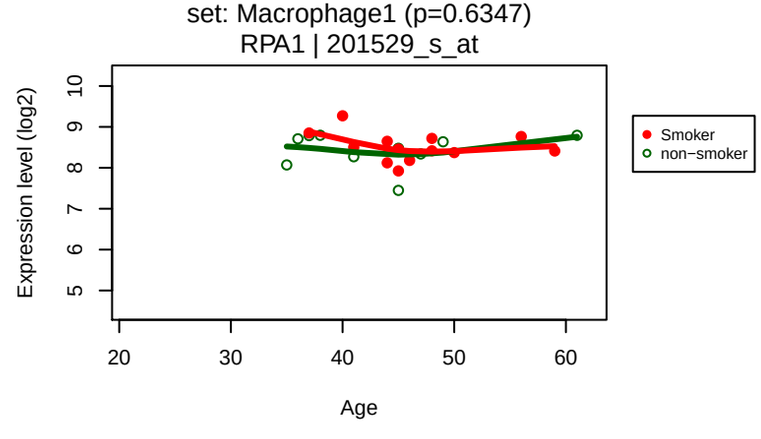

set: SmallAirway1 (p=0.7647)
RPA1 | 201529_s_at
Expression level (log2)
Age
Smoker
non−smoker

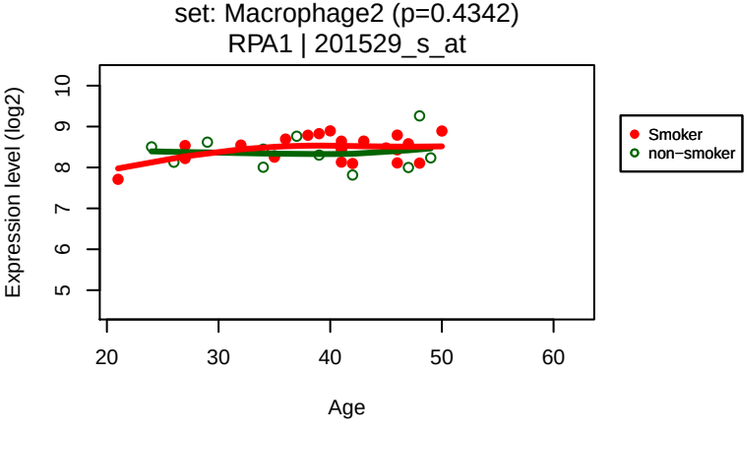

set: SmallAirway2 (p=0.4448)
RPA1 | 201529_s_at
Expression level (log2)
Age
Smoker
non−smoker

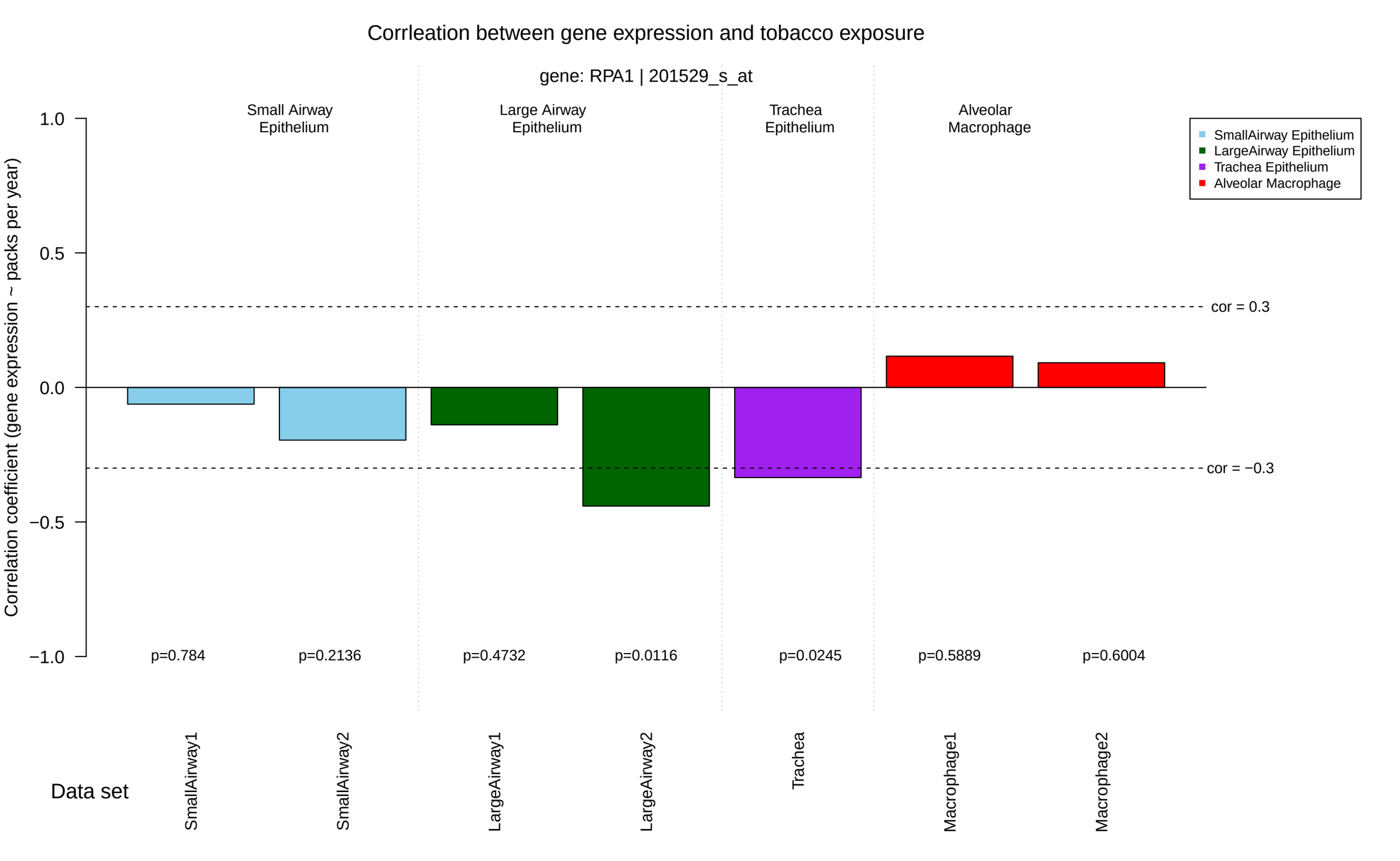

set: LargeAirway1 (p=0.0306)
RPA1 | 201529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0057)
RPA1 | 201529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.021)
RPA1 | 201529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6347)
RPA1 | 201529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4342)
RPA1 | 201529_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPA1 | 201529_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.784
p=0.2136
p=0.4732
p=0.0116
p=0.0245
p=0.5889
p=0.6004
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2